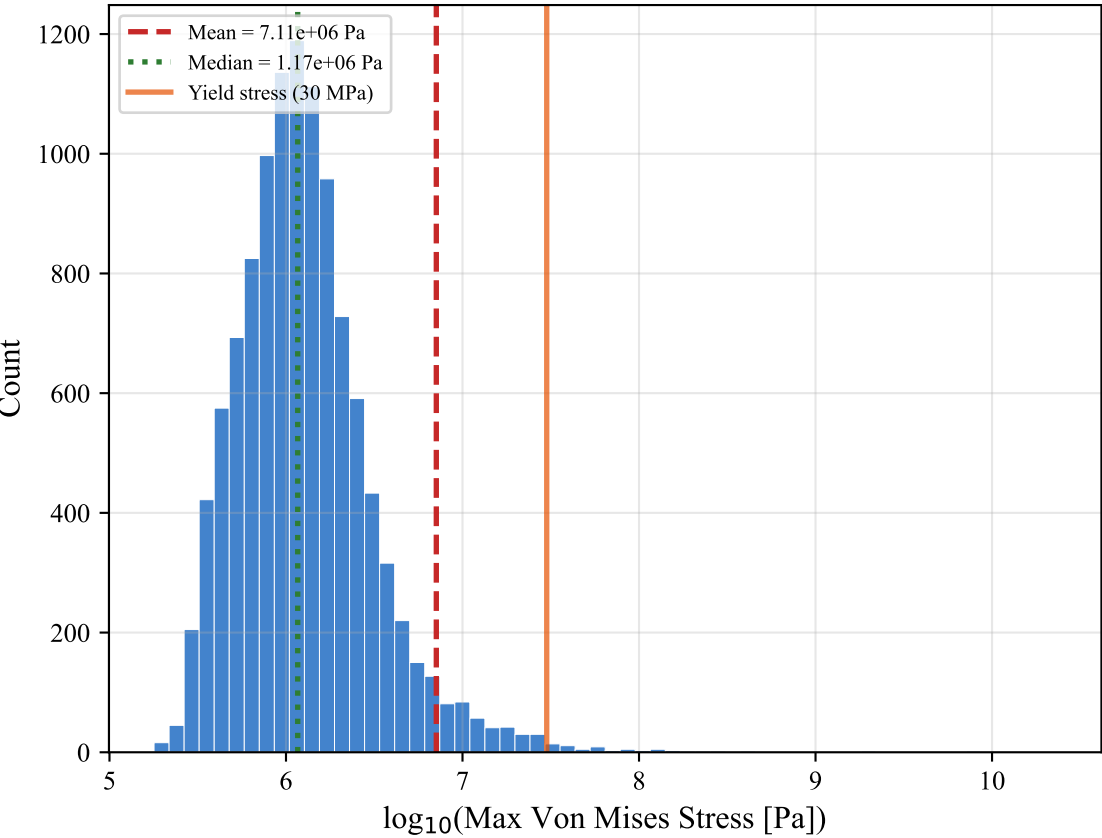
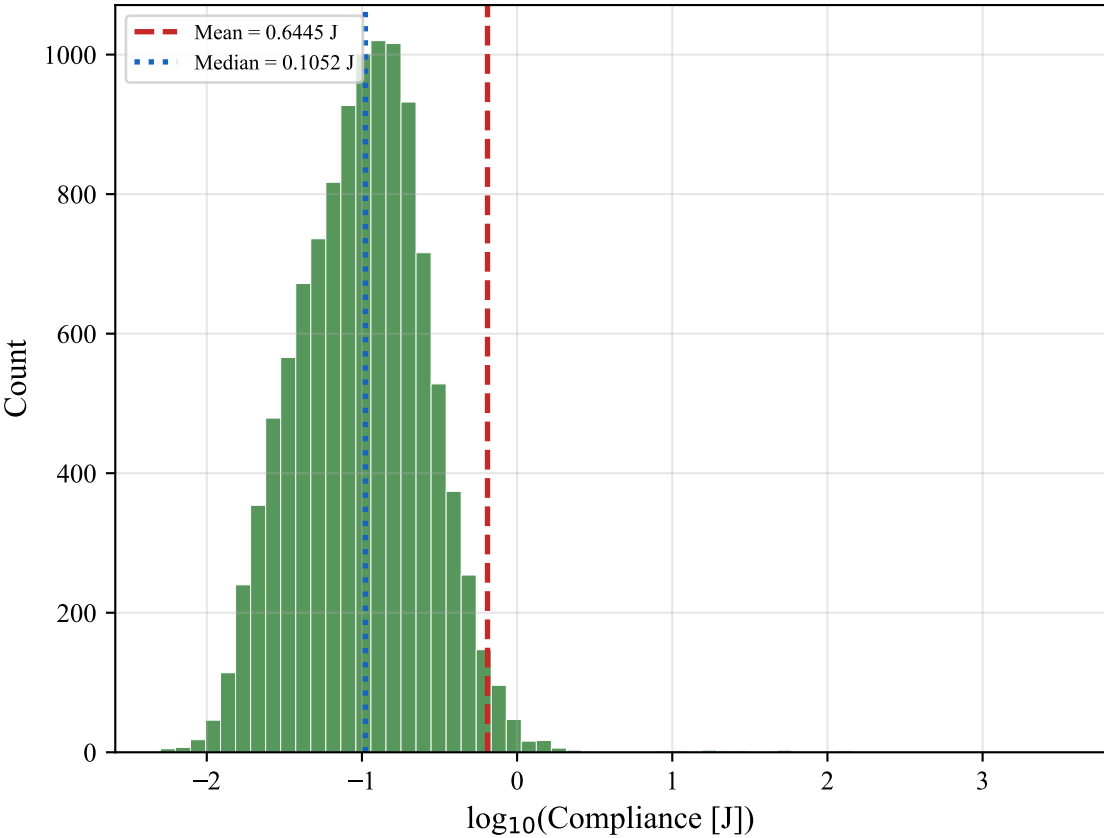


Figure 14: FEA Training Dataset Distributions (14,293 Simulations)

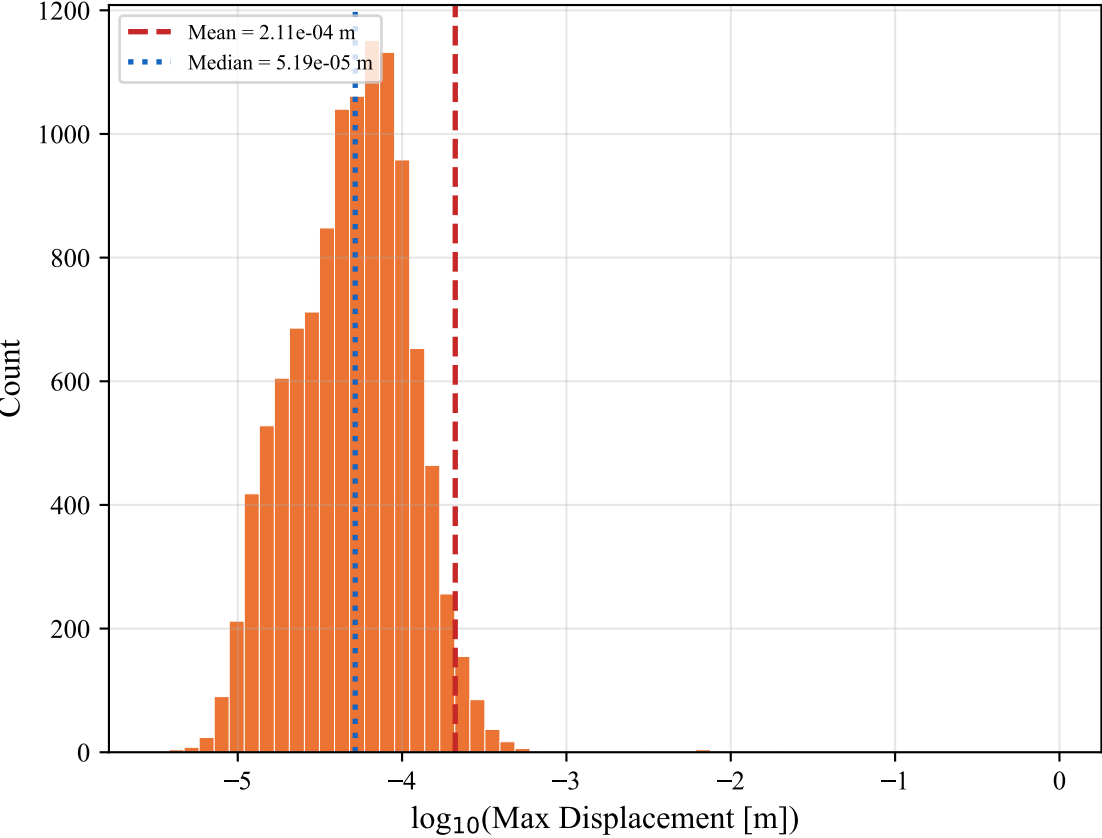
(a) Von Mises Stress Distribution



(b) Compliance Distribution



(c) Displacement Distribution



(d) Dataset Summary

Pipeline Statistics	
Total simulations	14,293
Clean samples	11,178
Rejected samples	3,115
Rejection rate	21.8%
Clean Data Statistics	
VM stress (median)	1.17×10^6 Pa
Compliance (median)	0.1052 J
Displacement (median)	5.19×10^{-5} m
Safety factor (median)	25.7×